



**RAJALAKSHMI
ENGINEERING
COLLEGE**

**DEPARTMENT OF INFORMATION
TECHNOLOGY LAB MANUAL**

CS23432 – Software Construction

(REGULATION 2023)

**RAJALAKSHMI ENGINEERING COLLEGE
Thandalam, Chennai-602015**

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Semester: IV

Academic Year: 2025-2026

INDEX

S.No.	Date	Title
1.	21/01/26	Azure Devops Environment Setup.
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3.	04/02/26	Setting Up Epics, Features, And User Stories for Project Planning.
4.	11/02/26	Sprint Planning.
5.	18/02/26	Poker Estimation.
6.	25/02/26	Designing Class and Sequence Diagrams for Project Architecture.
7.	11/03/26	Designing Architectural and ER Diagrams for Project Structure.
8.	18/03/26	Testing – Test Plans and Test Cases.
9.	25/03/26	Load Testing and Pipelines.
10.	08/04/26	GitHub: Project Structure & Naming Conventions.

EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP
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Aim:

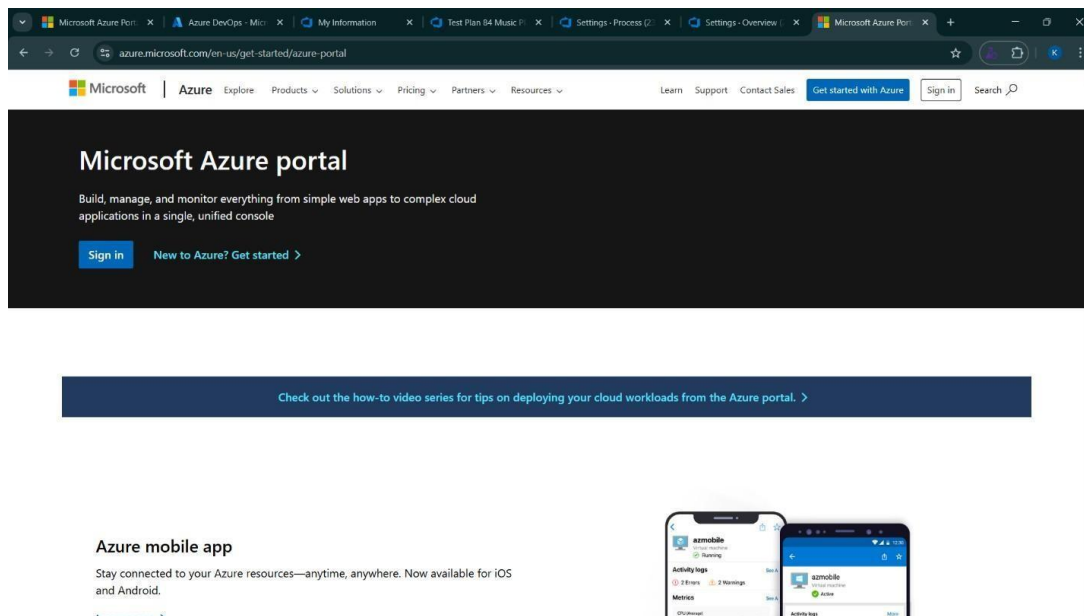
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

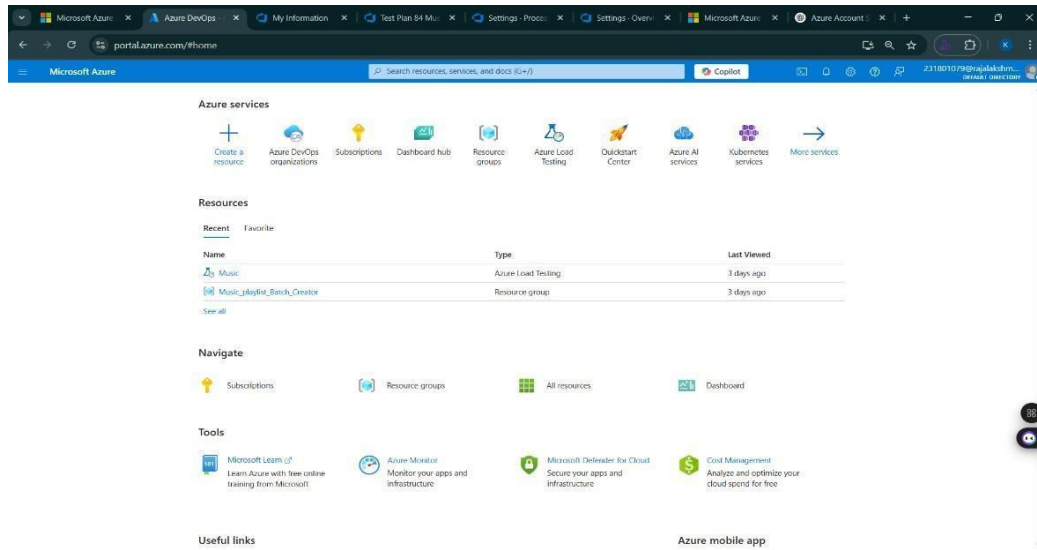
1. Open your web browser and go to the Azure website:
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

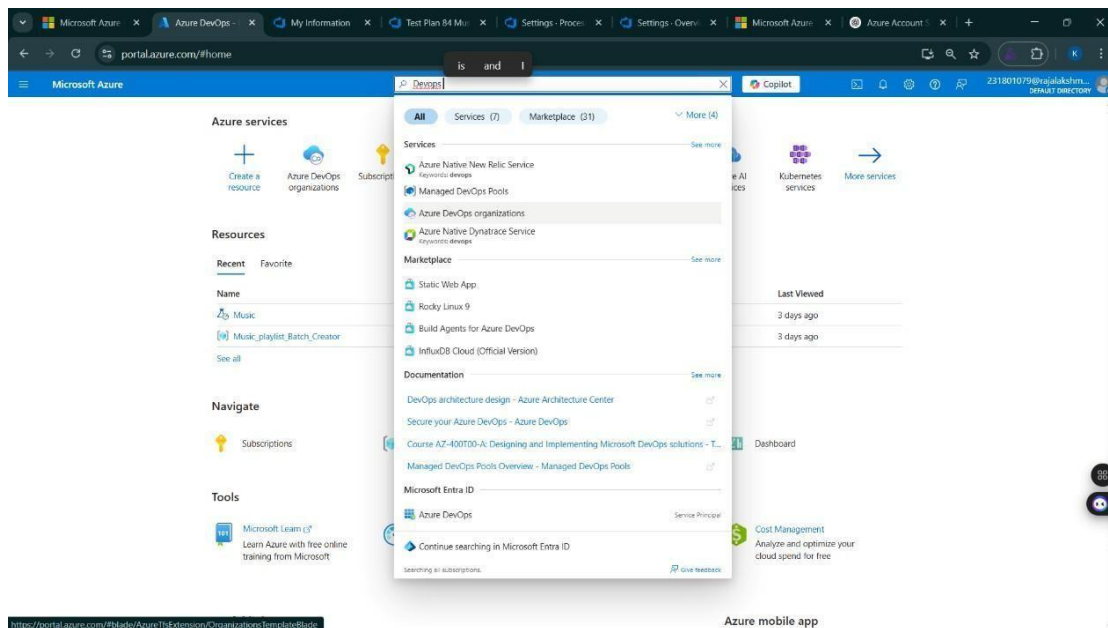
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



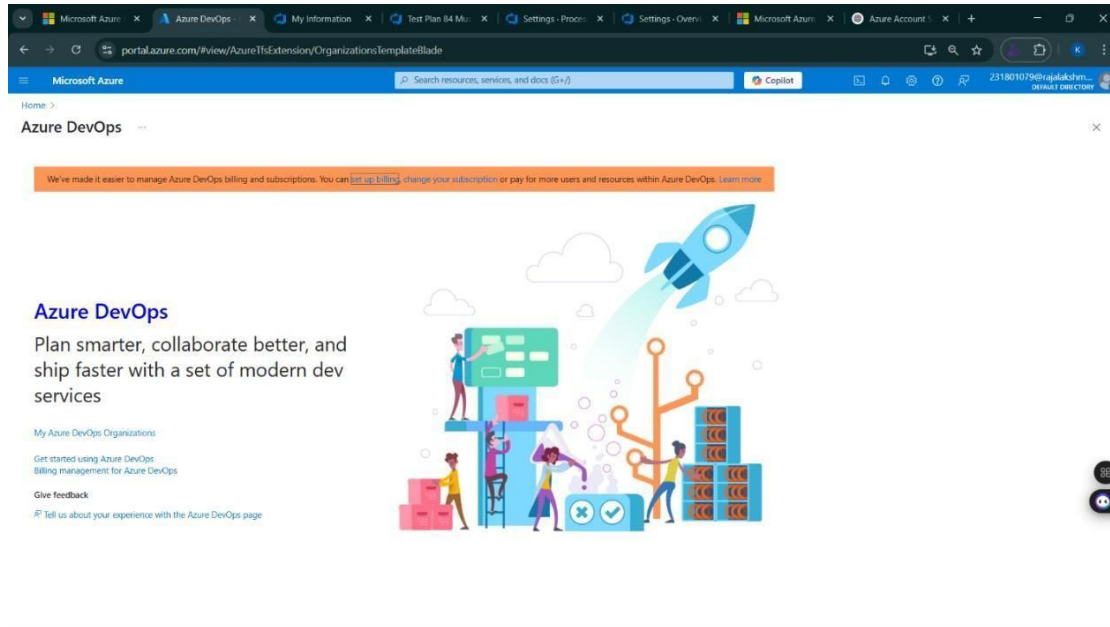
2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the *My Azure DevOps Organization* link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

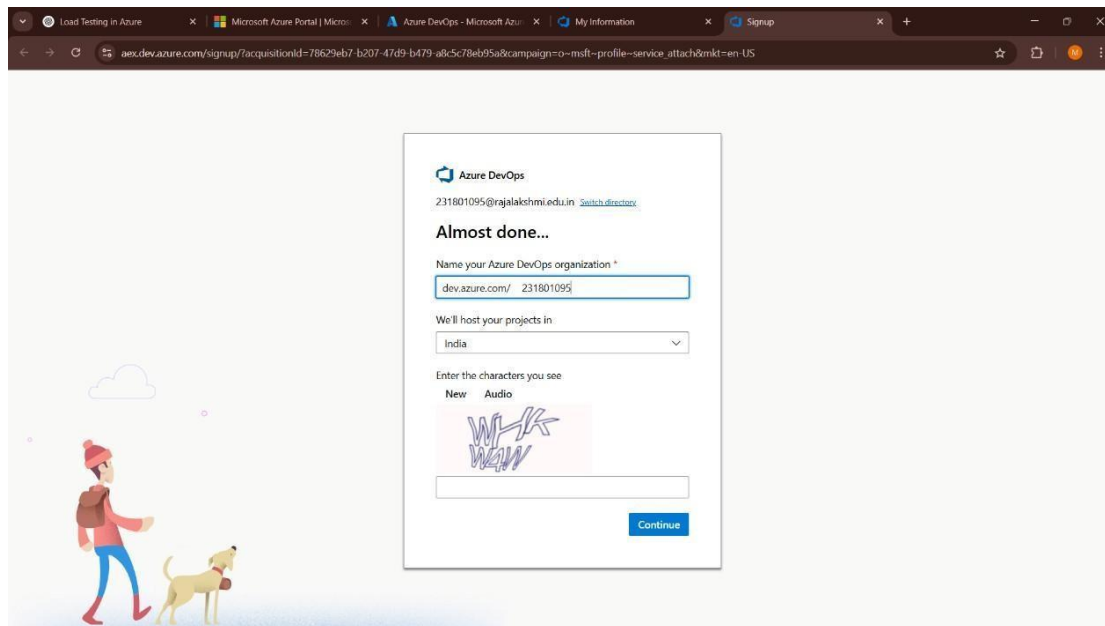
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
-----------	---

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account



2. Create the First Project in Your Organization

a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's **Home page**, click on the **New Project** button.

c. Enter the project name, description, and visibility options:

Name: Choose a name for the project (e.g., **To-Do List App**).

Description: Optionally, add a description to provide more context about the project.

Visibility: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

d. Once you've filled out the details, click **Create** to set up your first project.

Create new project



Project name *

TO DO LIST

Description

Visibility



Public

Anyone on the internet can view the project. Certain features like TFVC are not supported.



Private

Only people you give access to will be able to view this project.

By creating this project, you agree to the Azure DevOps [code of conduct](#)

^ Advanced

Version control ?

Git



Work item process ?

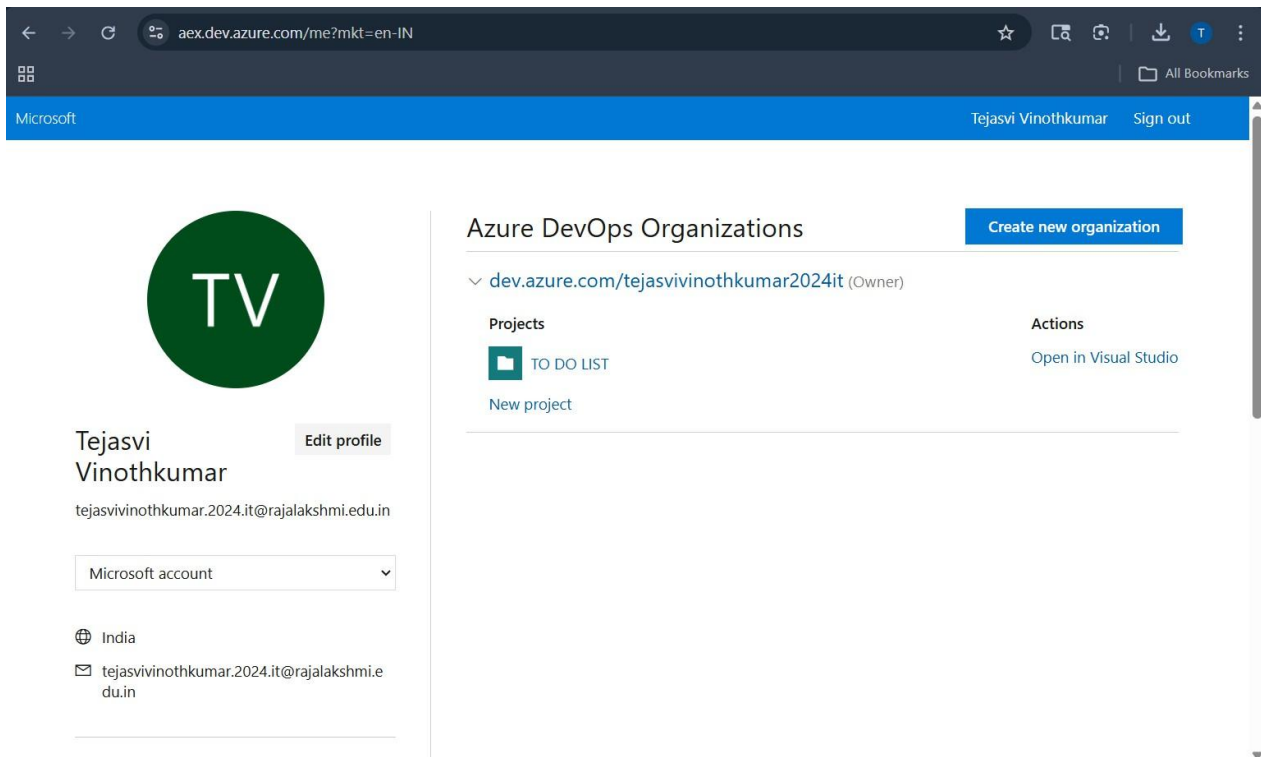
231801095 Agile



Cancel

Create

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



4. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

tejasvinothkumar2024it / TO DO LIST / Boards / Backlogs

Search

TO DO LIST Team

+ New Work Item View as Board Column Options

Backlog Analytics

Epics

	Order	Work Item Type	Title	State	Effort	Busin...	Value Area	Tags
1	Epic	>	Task Management	New			Business	
2	Epic		Scheduled Reminders	New			Business	
3	Epic	>	Performance Requirement	New			Business	
4	Epic		Reliability Requirement	New			Business	
5	Epic		Security Requirement	New			Business	

Result:

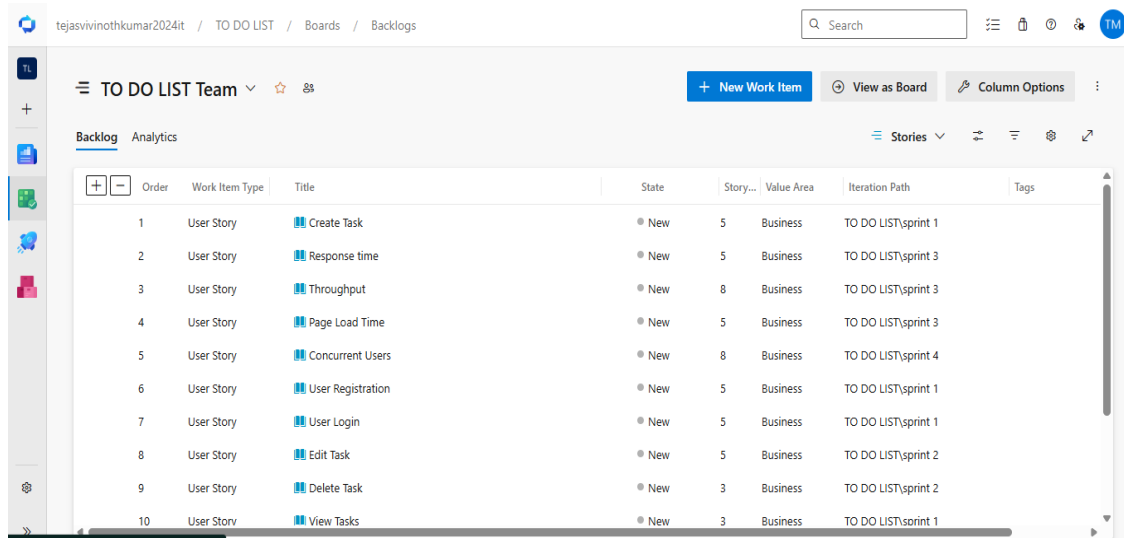
Successfully created an Azure DevOps project with user story management and agile workflow setup.

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task



The screenshot shows the Jira interface for the 'TO DO LIST Team'. The 'Backlog' tab is selected, displaying a list of 10 user stories. Each story is represented by a card with a title, state, story key, value area, iteration path, and tags. The stories are numbered 1 through 10 and all have a state of 'New'.

Order	Work Item Type	Title	State	Story...	Value Area	Iteration Path	Tags
1	User Story	Create Task	New	5	Business	TO DO LIST\sprint 1	
2	User Story	Response time	New	5	Business	TO DO LIST\sprint 3	
3	User Story	Throughput	New	8	Business	TO DO LIST\sprint 3	
4	User Story	Page Load Time	New	5	Business	TO DO LIST\sprint 3	
5	User Story	Concurrent Users	New	8	Business	TO DO LIST\sprint 4	
6	User Story	User Registration	New	5	Business	TO DO LIST\sprint 1	
7	User Story	User Login	New	5	Business	TO DO LIST\sprint 1	
8	User Story	Edit Task	New	5	Business	TO DO LIST\sprint 2	
9	User Story	Delete Task	New	3	Business	TO DO LIST\sprint 2	
10	User Story	View Tasks	New	3	Business	TO DO LIST\sprint 1	

1.Fill in Epics

3.Fill in User Story Details

USER STORY 14

14 Throughput

Yuvashree D

0 Comments Add Tag

Save and Close

Follow

Updated by Tejasvi Vinothkuman 1h ag

Details 1 2

Description

Acceptance Criteria

The system must support at least 50 task operations per minute .Task operations include create,update,delete, and view tasks.

Discussion

Planning

Story Points
8

Priority
2

Risk

Classification

Value area
Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Result:

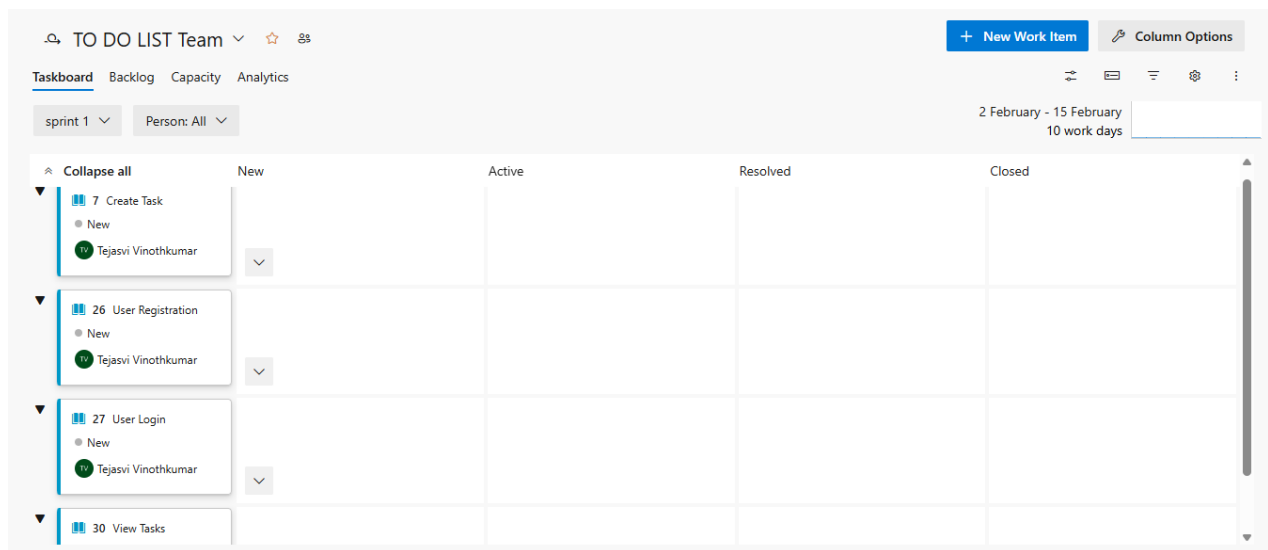
Thus, the creation of epics, features, user story and task has been created successfully.

SPRINT PLANNING

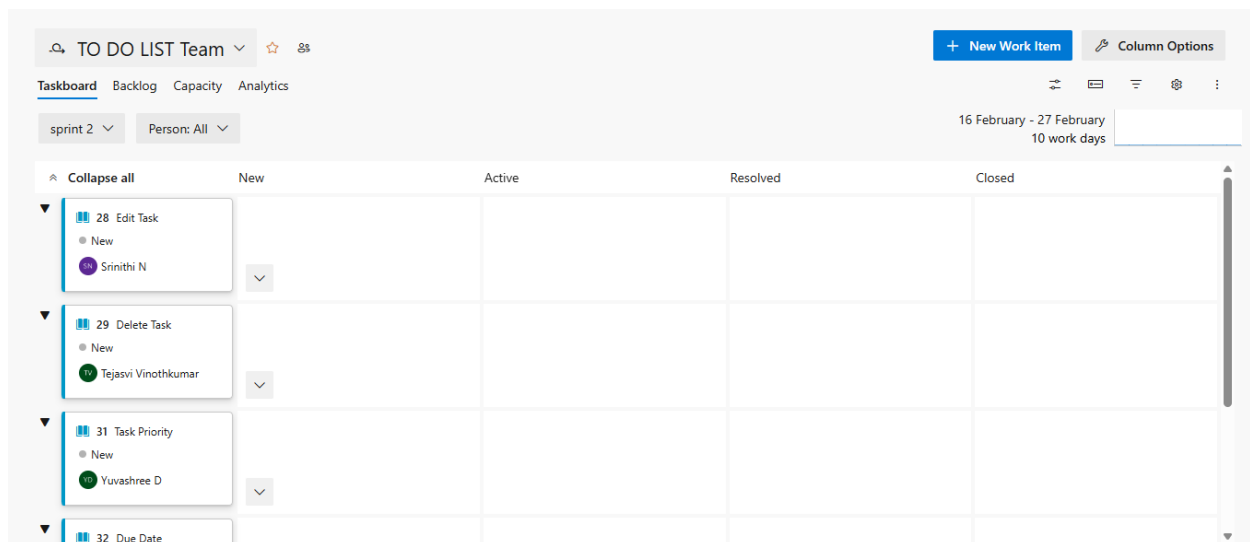
Aim:

To assign user story to specific sprint for the Advanced Reliable To-Do List Application.

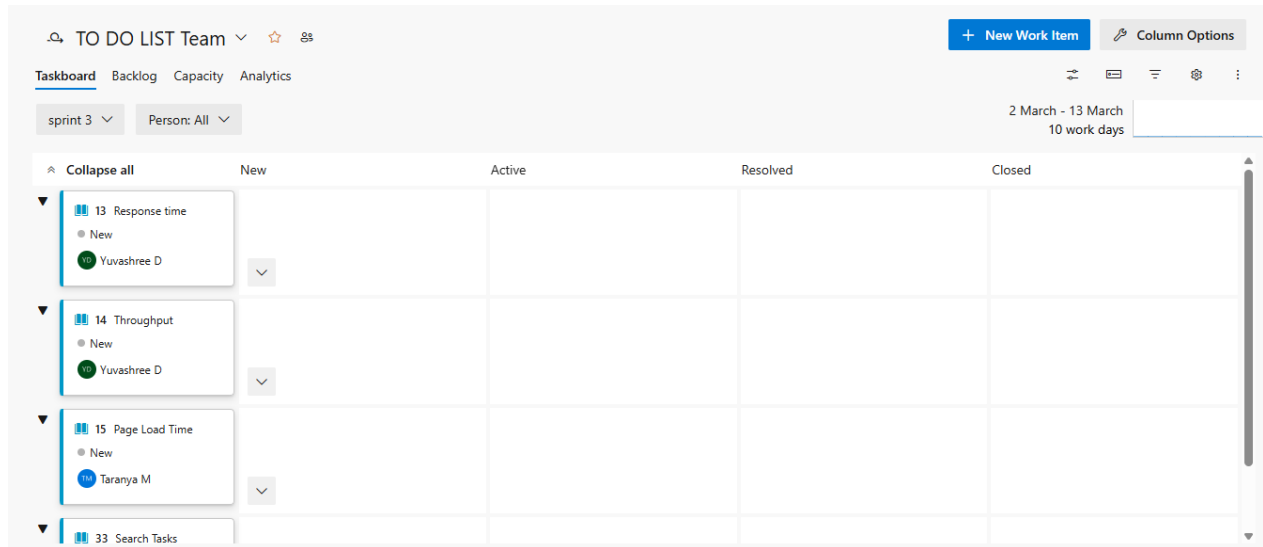
Sprint Planning Sprint 1



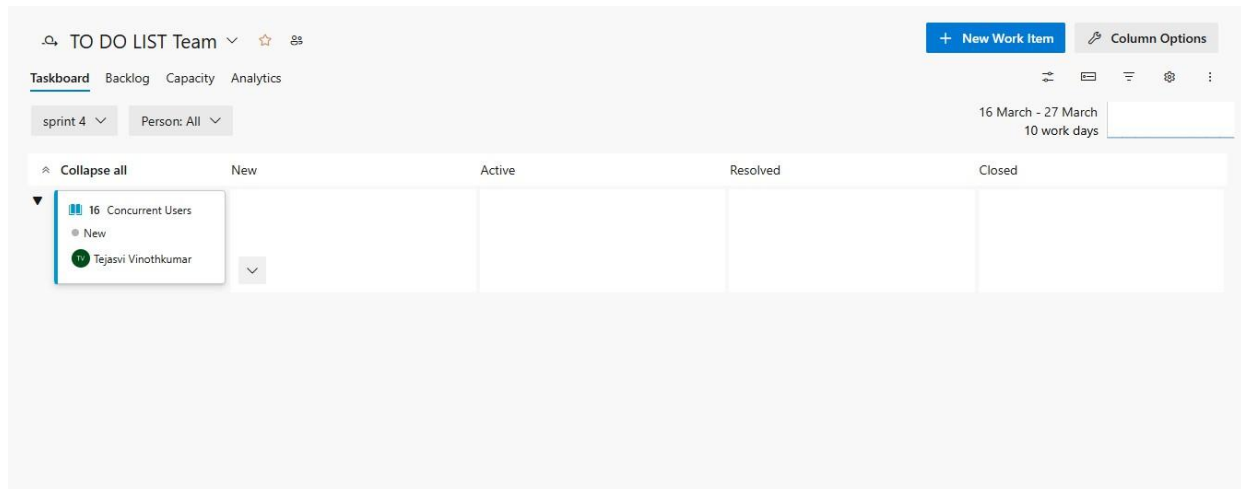
Sprint 2



Sprint 3



Sprint 4



Result:

The Sprints are created for the To-Do List Application.

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories - Music Playlist Batch Creator Project.

Poker Estimation

USER STORY 34

34 Mark Task as Completed

Yuvashree D

0 Comments Add Tag

Save and Close

Follow

State: New

Area: TO DO LIST

Reason: New

Iteration: TO DO LIST\sprint 2

Updated by Tejasvi Vinothkumar: 2h ago

Details

1

2

Description

Acceptance Criteria

- User can change task status to completed.
- Completed tasks are visually differentiated.
- Status update is saved successfully.
- Completed tasks remain visible in task history.

Discussion

Planning

Story Points: 3

Priority: 2

Risk

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Result:

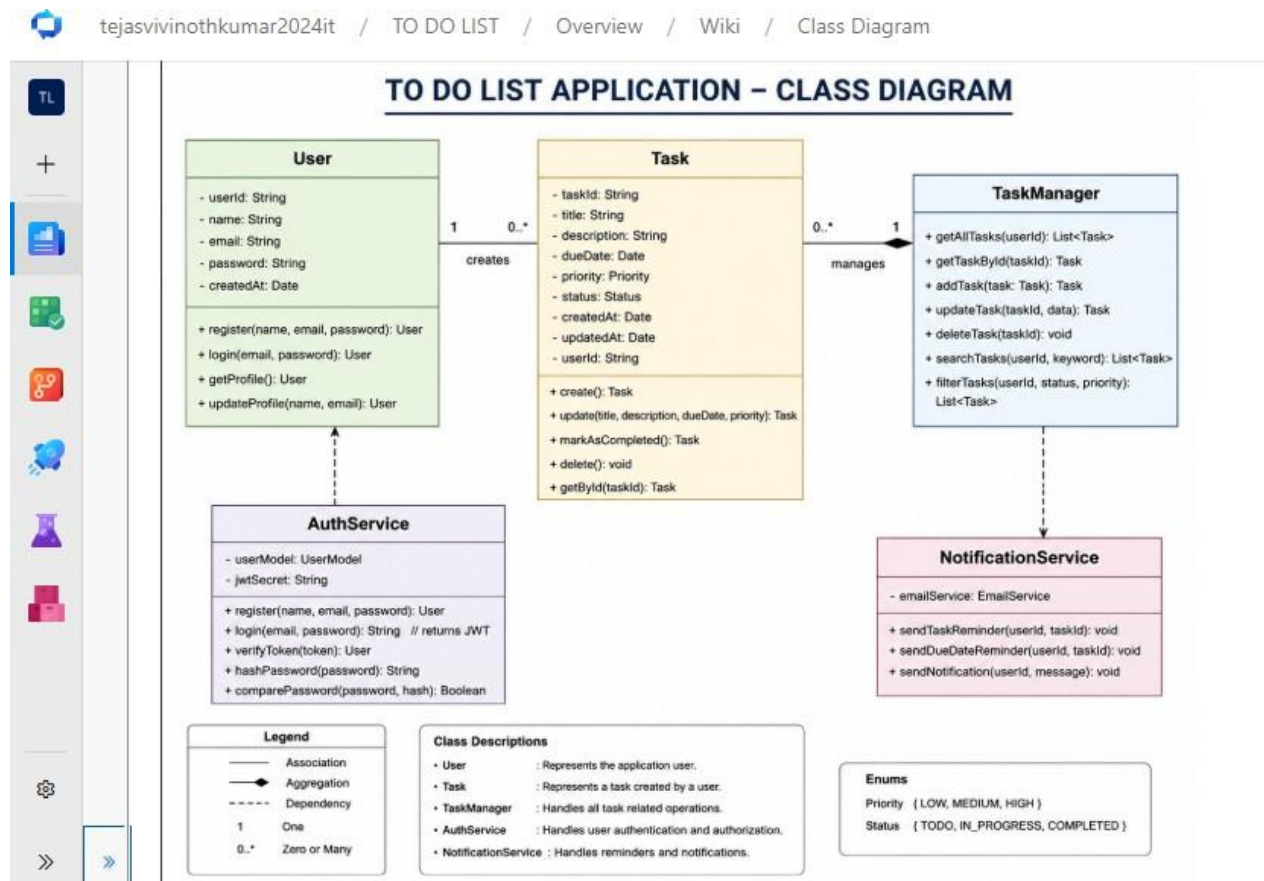
The Estimation/Story Points is created for the project using Poker Estimation.

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

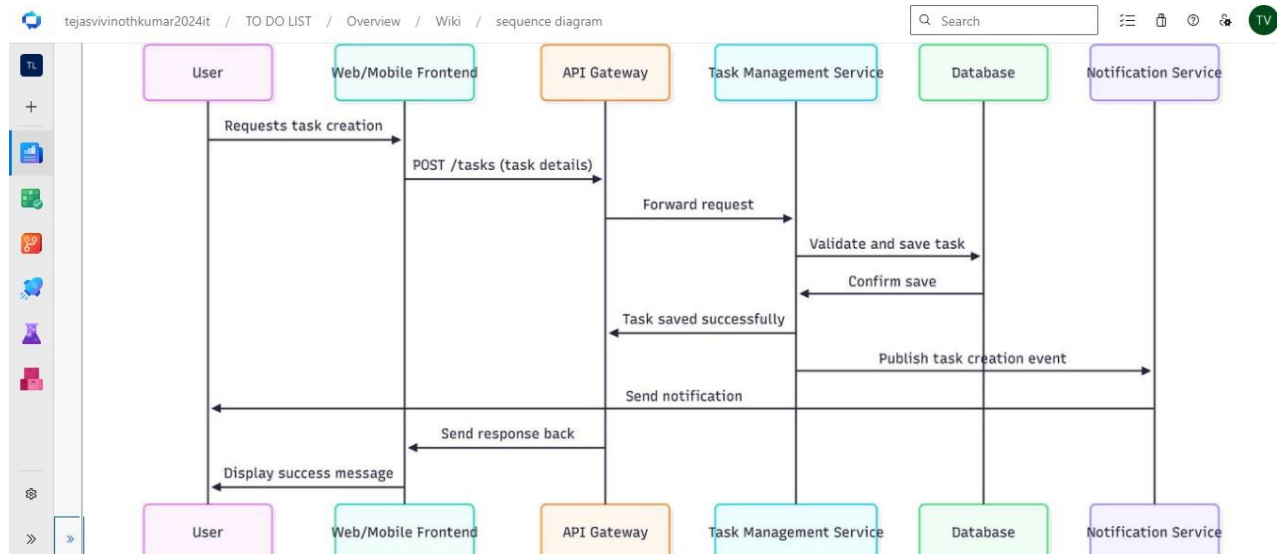
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project. **6A.**

Class Diagram



6B. Sequence Diagram



Result:

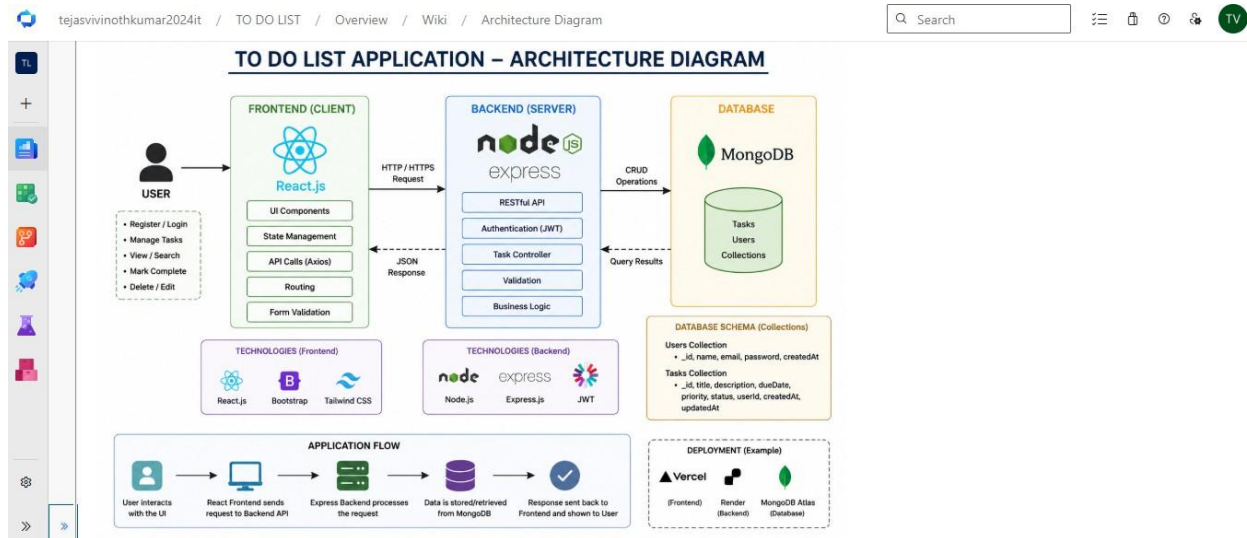
The Class Diagram and Sequence Diagram is designed Successfully for the To-Do List Application.

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

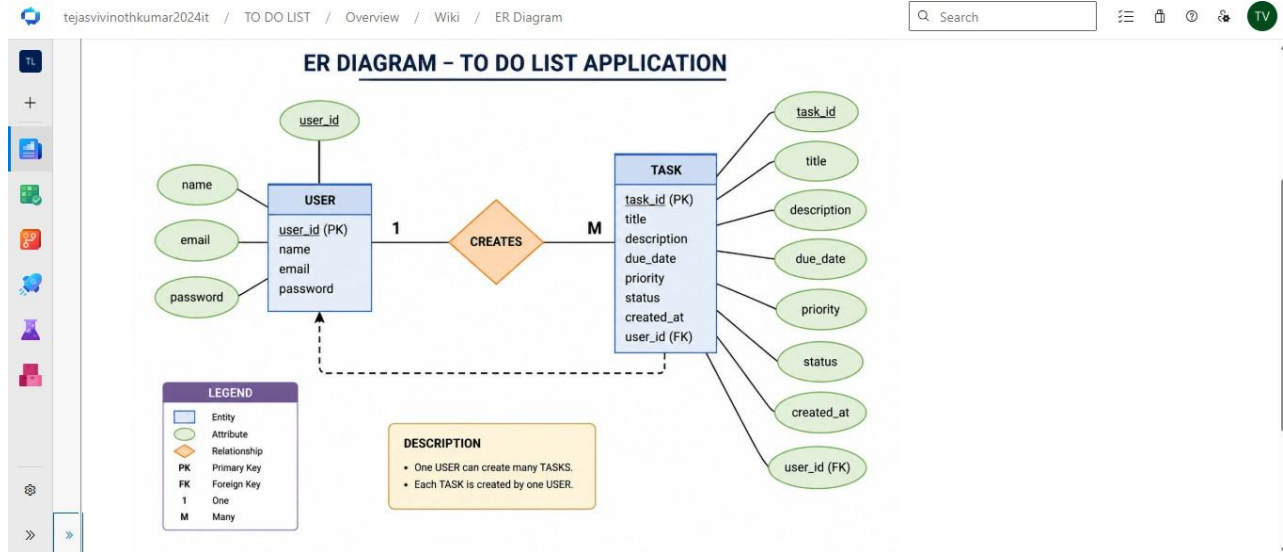
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the To-Do List Application.

	TESTING – TEST PLANS AND TEST CASES
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Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1) Understand Core Features of the Application

- User Registration & Login
- Create, Update, Delete Tasks
- Task Prioritization & Filtering
- Mark Task as Completed
- Notifications / Reminders
- Data Sync with Cloud (Azure)

2) Define User Interactions

Each test case simulates real user behavior

Eg: logging in, adding tasks, editing tasks, marking complete.

3) Design Happy Path Test Cases

Validates that all features work under normal conditions

Eg: User logs in → creates task → marks it complete successfully

4) Design Error Path Test Cases

Tests invalid or unexpected scenarios

Eg: login fails with wrong password, task creation fails with empty input

5) Break Down Steps and Expected Results

Each test case includes:

Step-by-step actions

Expected output

6) Use Clear Naming and IDs

Eg:

TC01 – Successful Login

TC02 – Login Failure

7) Separate Test Suites

Group test cases based on functionality:

(Login, Task Management, Notifications)

8) Prioritize and Review

Critical features → High priority

Ensure completeness and accuracy

1. New test plan

tejasvinothkumar2024it / TO DO LIST / Test Plans

Search

New Test Plan

Name *

To-Do List - Test Plan

Area Path *

TO DO LIST

Iteration *

TO DO LIST\sprint 1 02-02-2026 - 15-02-2026

Create Cancel

2. Test suite

tejasvinothkumar2024it / TO DO LIST / Test Plans / TO DO LIST

Search

edit tasks (ID: 44)

Define Execute Chart

Test Suites

Filter suites by name

TO DO LIST

- edit tasks (1)
- search tasks (1)
- view tasks (1)
- create task (1)

Test Points (1 item)

Title	Outcome	Order	Test Case Id	State
edit task	Passed	1	45	Design

3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

To-Do List Application – Test Plans

USER STORIES

- As a user, I want to register and log in so that I can manage my tasks
- As a user, I want to create and organize tasks
- As a user, I want to update or delete tasks
- As a user, I want to mark tasks as completed
- As a user, I want to receive reminders/notifications

Test Suites

Test Suite: TS01 – User Login (ID: 101)

1) TC01 – Successful Sign Up

Action:

- Go to Sign-Up page
- Enter valid details
- Click "Sign Up"

Expected Results:

- Account created successfully
- Redirected to dashboard
- Type: Happy Path

2) TC02 – Secure Login

Action:

- Go to Login page
- Enter valid credentials
- Click "Login"
- Expected Results:
- Login successful
- Dashboard displayed
- Type: Happy Path

3) TC03 – Sign Up with Existing Email

Action:

- Enter already registered email
- Click "Sign Up"
- Expected Results:

- Error message: "Email already exists"
- Type: Error Path

4) TC04 – Login with Wrong Password

Action:

- Enter incorrect password
- Click "Login"
- Expected Results:
- Error message displayed
- Type: Error Path

Test Suite: TS02 – Task Management (ID: 102)

1) TC05 – View Task List

Action:

- Login
- Open task dashboard
- Expected Results:
- List of tasks displayed
- Type: Happy Path

2) TC06 – Task Load Failure (No Internet)

Action:

- Turn off internet
- Open task list
- Expected Results:
- Error message displayed
- Type: Error Path

Test Suite: TS03 – Task Operations (ID: 103)

1) TC07 – Add Task

Action:

- Enter task details
- Click "Add Task"
- Expected Results:

- Task added successfully
- Type: Happy Path

2) TC08 – Add Task with Missing Data

Action:

- Leave fields empty
- Submit
- Expected Results:
- Validation error shown
- Type: Error Path

Test Suite: TS04 – Task Completion & Review (ID: 104)

1) TC09 – Mark Task as Completed

Action:

- Select a task
- Click "Mark as Done"
- Expected Results:
- Task marked as completed
- Type: Happy Path

2) TC10 – Mark Non-existing Task

Action:

- Try marking invalid/non-existing task
- Expected Results:
- Error message shown
- Type: Error Path

Test Suite: TS05 – Notifications (ID: 105)

1) TC11 – Receive Notification

Action:

- Set reminder for task
- Expected Results:
- Notification received via app/email
- Type: Happy Path

2) TC12 – Notification Failure

Action:

- Trigger notification with network issue
- Expected Results:
- Notification not sent / error logged
- Type: Error Path

Test Cases

TEST CASE 38

38 create task

Tejasvi Vinothkumar

0 Comments Add Tag

Save and Close Follow

State Design Area TO DO LIST Reason New Iteration TO DO LIST\sprint 1

Updated by Tejasvi Vinothkumar: 2h ago

Steps Summary Associated Automation 1 0

Steps

enter task title

task entered successfully

enter description

description entered successfully

enter due date

due date entered successfully

enter priority

priority entered successfully

Click or type here to add a step

Recent test results

Passed by with Windows 10

Completed 09:20, 10-05-2026

Passed by with Windows 10

Completed 09:20, 10-05-2026

Passed by with Windows 10

Completed 09:18, 10-05-2026

View all test result history

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for

TEST CASE 43

43 search tasks

Tejasvi Vinothkumar

0 Comments Add Tag

Save and Close Follow

State Design Area TO DO LIST Reason New Iteration TO DO LIST\sprint 1

Updated by Tejasvi Vinothkumar: 2h ago

Steps Summary Associated Automation 1 0

Steps

Search existing task

Matching tasks should be filtered correctly.

Search partial keyword

Matching tasks should be filtered correctly.

Search non-existing task

Matching tasks should be filtered correctly.

Click or type here to add a step

Recent test results

Passed by with Windows 10

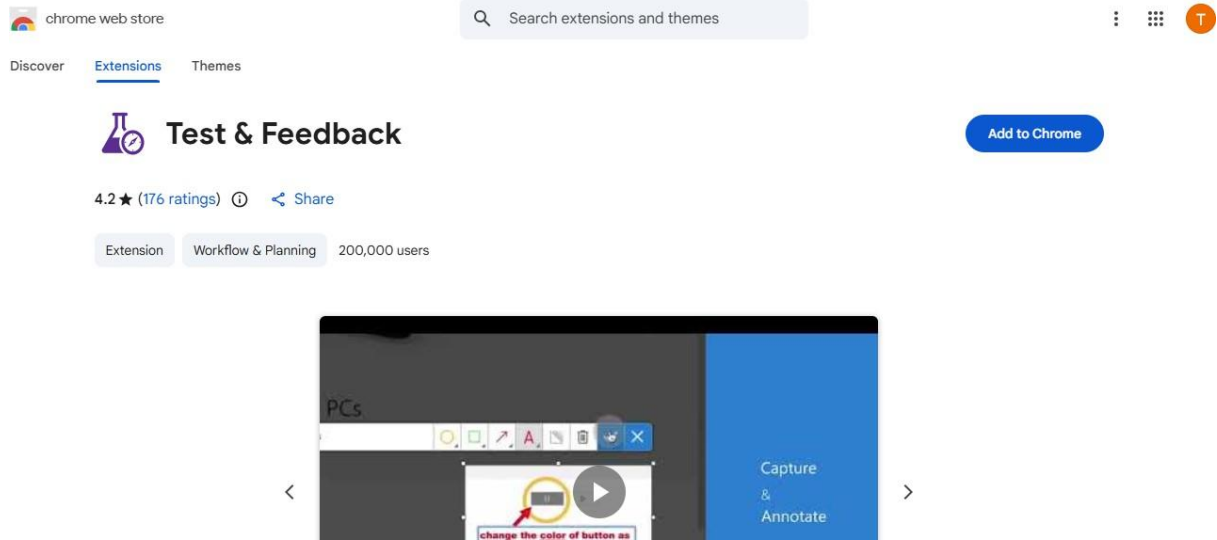
Completed 09:28, 10-05-2026

Deployment

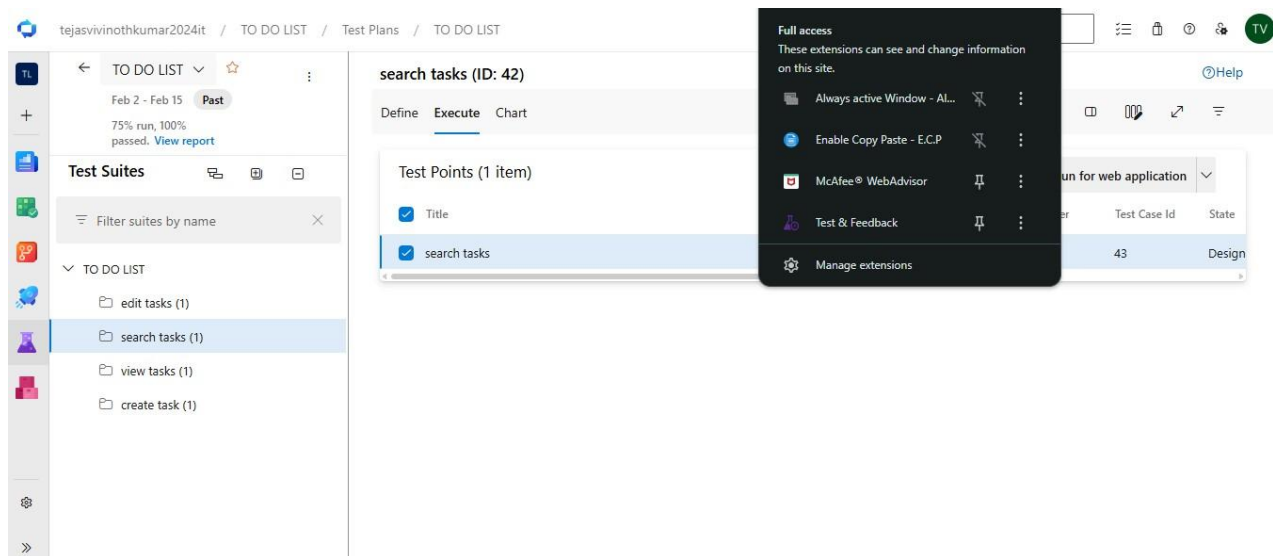
To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

4. Installation of test



Test and feedback
Showing it as an extension



5. Running the test cases

tejasvinothkumar2024it / TO DO LIST / Test Plans / TO DO LIST

Search

edit tasks (ID: 44)

Define Execute Chart

Test Suites

Filter suites by name

TO DO LIST

- edit tasks (1)
- search tasks (1)
- view tasks (1)
- create task (1)

Test Points (1 item)

Title	Outcome	Order	Test Case Id	State
edit task	Passed	1	45	Design

Run for web application

- View execution history
- Mark Outcome
- Run
- Reset test to active
- Edit test case
- Assign tester
- View test result

Run for web application

Run for desktop application

Run with options

Runner - Test Plans - Google Chrome

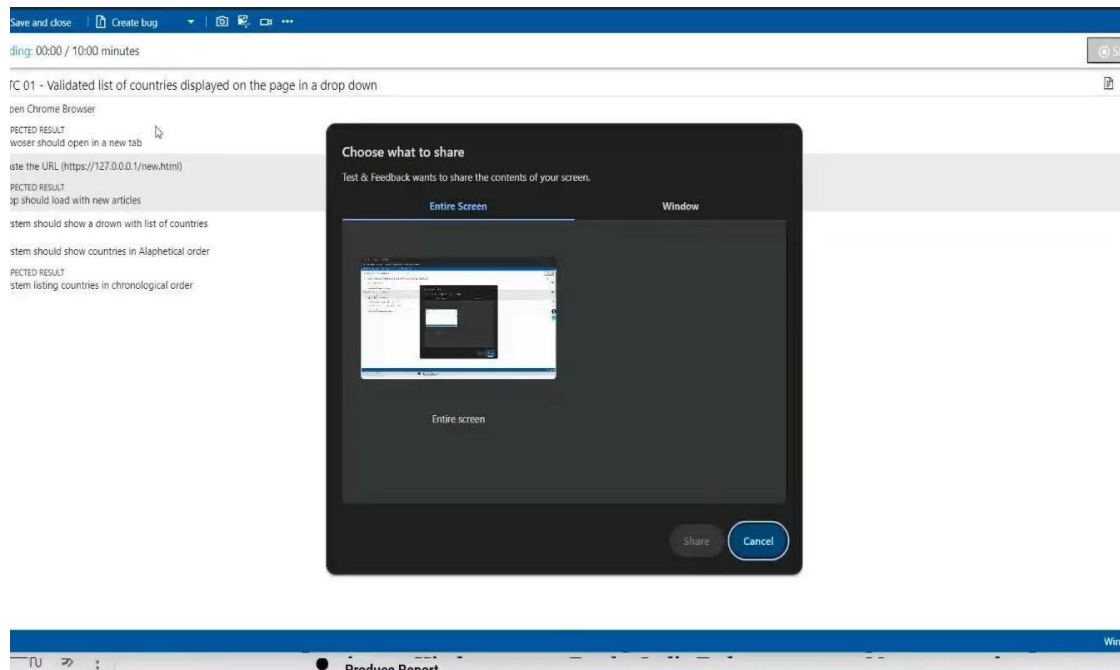
dev.azure.com/tejasvinothkumar2024it/TO%20DO%20LIST/_testExecution/Index

Save and close... Create bug

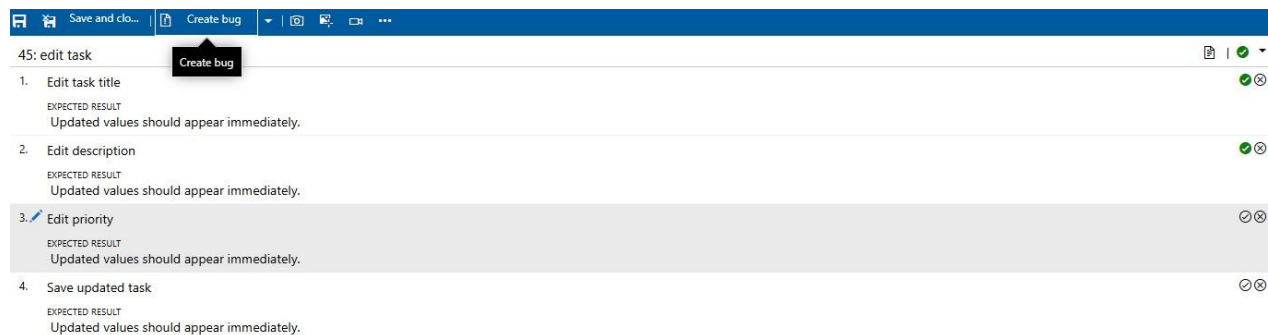
45: edit task

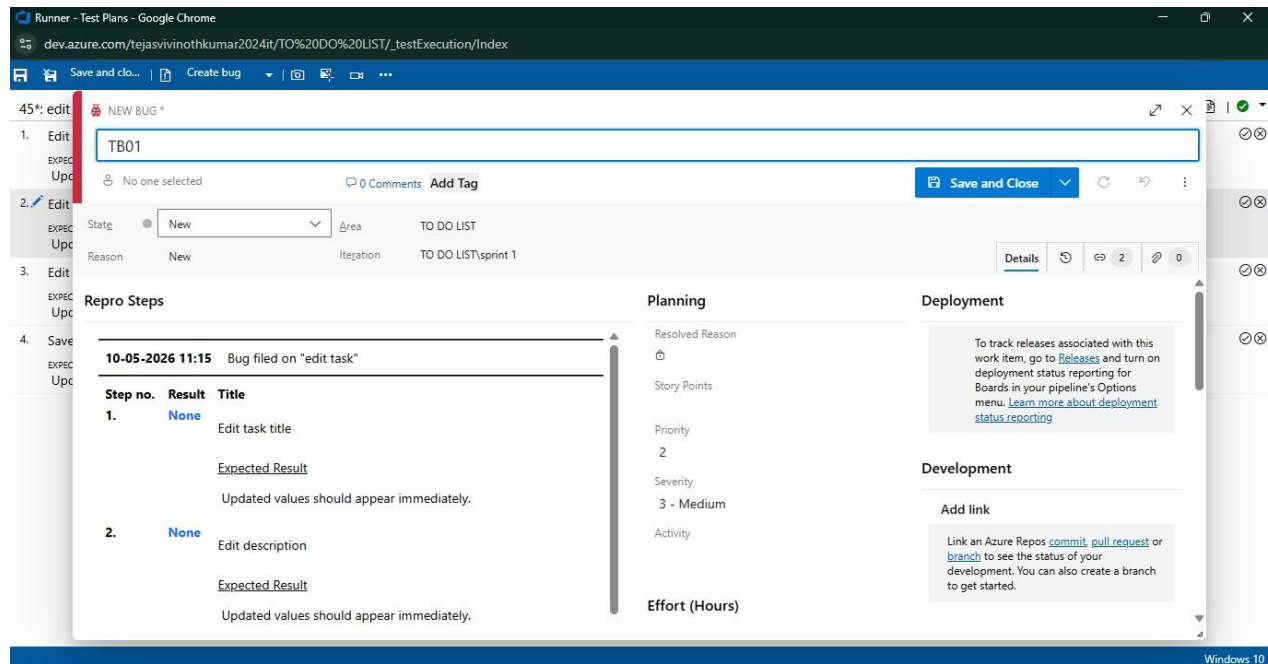
- Edit task title
EXPECTED RESULT
Updated values should appear immediately.
- Edit description
EXPECTED RESULT
Updated values should appear immediately.
- Edit priority
EXPECTED RESULT
Updated values should appear immediately.
- Save updated task
EXPECTED RESULT
Updated values should appear immediately.

6. Recording the test case

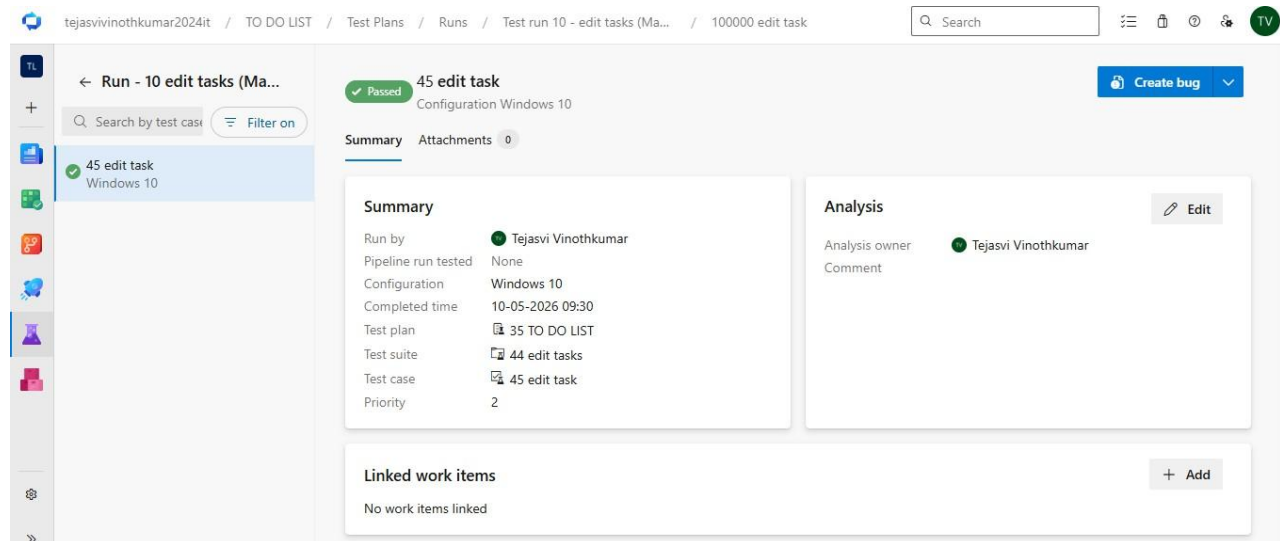


7. Creating the bug

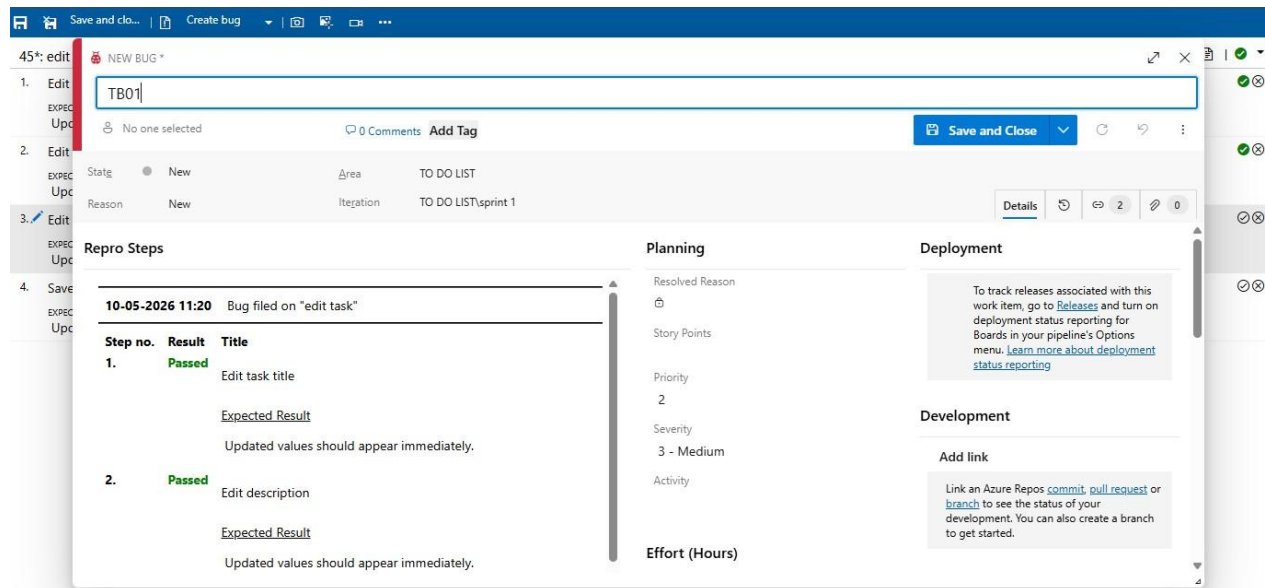




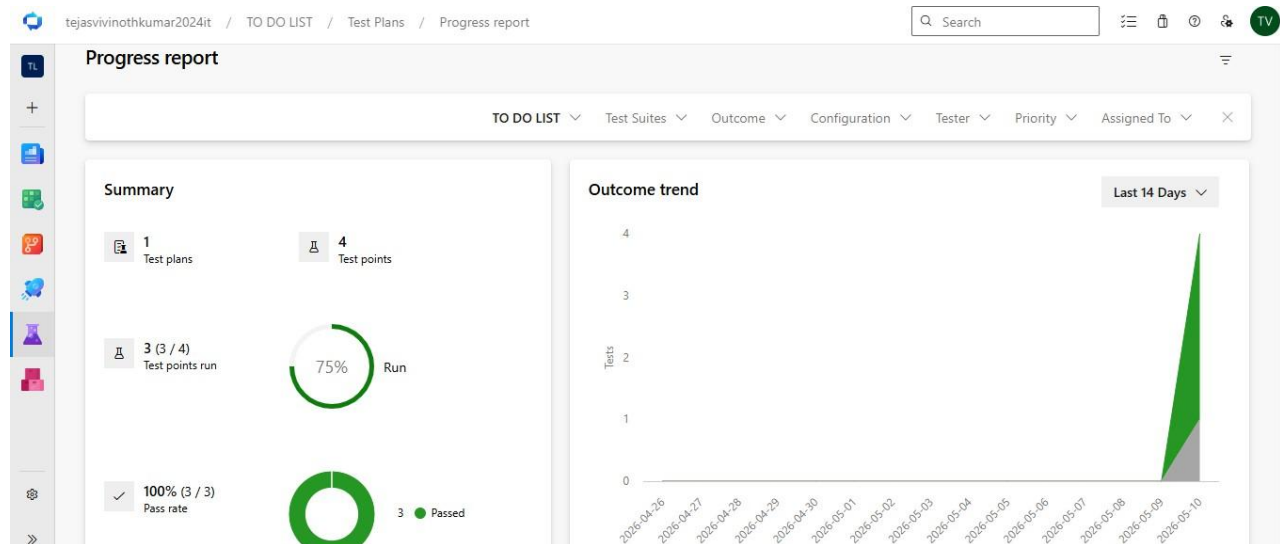
8. Test case results



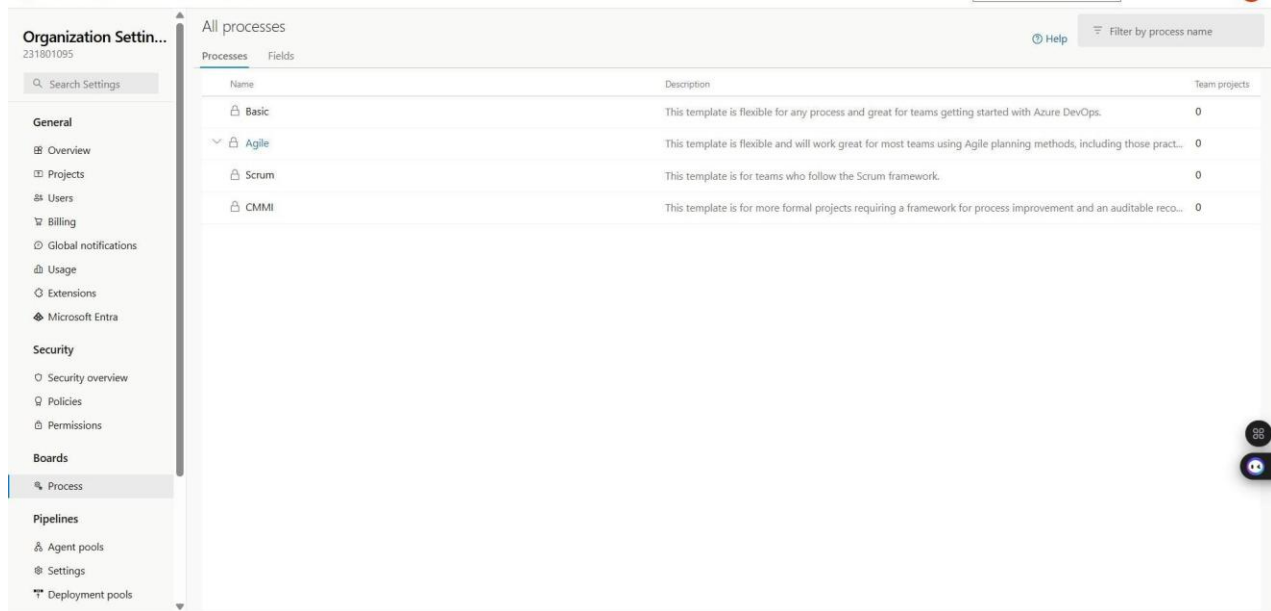
9. Test report summary



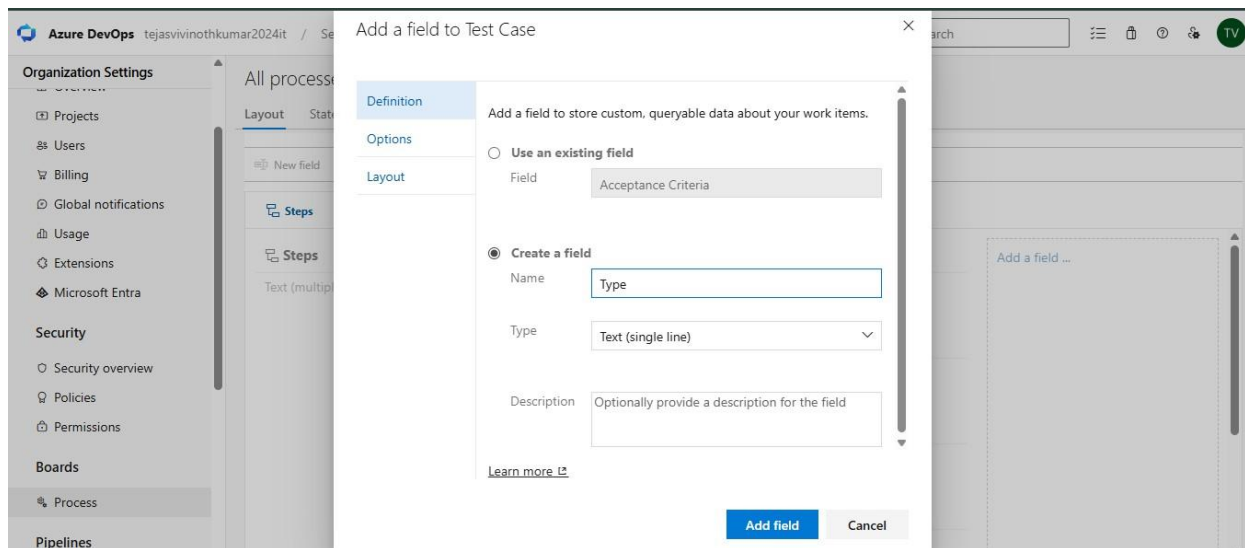
10. Progress report

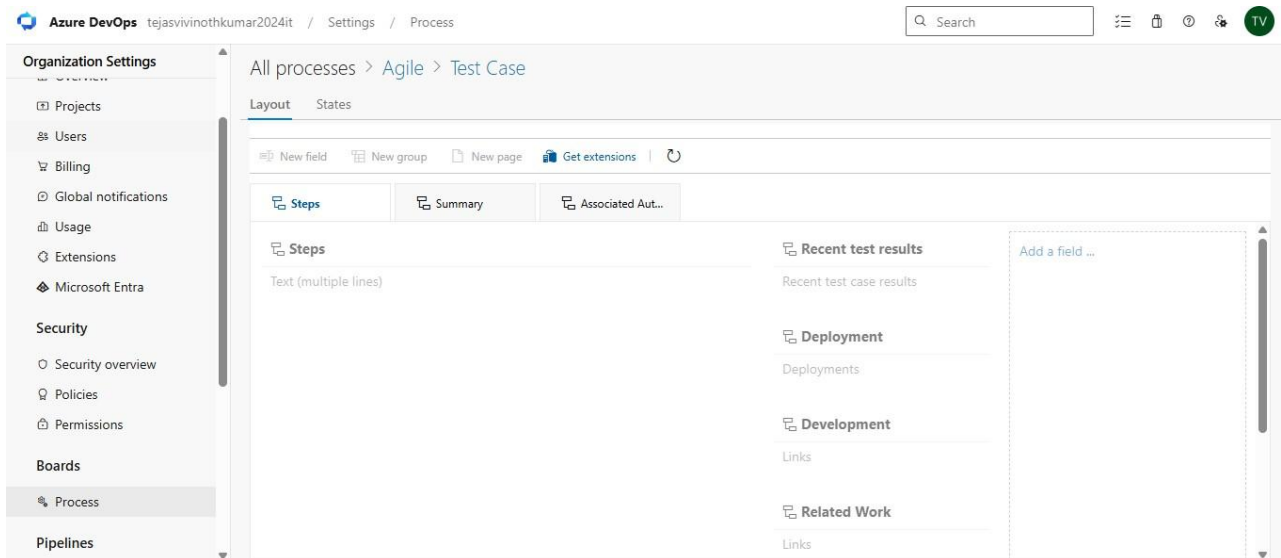
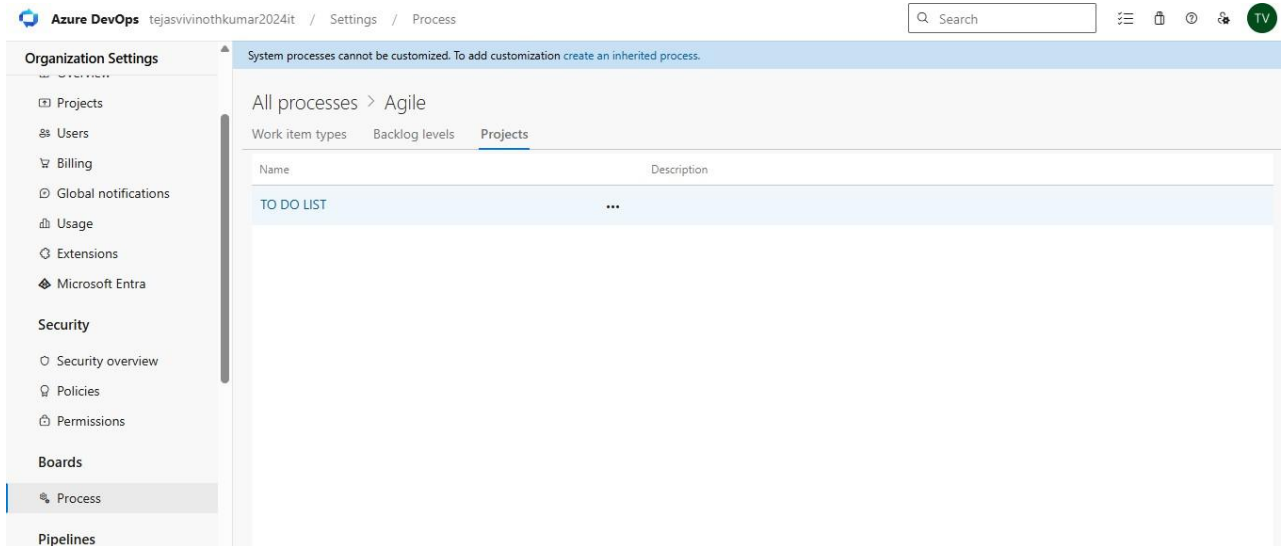


11. Changing the test template



12. View the new test case template





Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

	LOAD TESTING AND PIPELINES
--	-------------------------------------

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource o Go to *Create a resource* → Search for “Azure Load Testing”.
o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details o *Subscription*: Choose your Azure subscription. o *Resource Group*: Create new or select an existing one. o *Name*: Provide a unique name (no special characters).
o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab o *Test Name*: Provide a unique name.
o *Description*: (Optional) Add test purpose.
o *Run After Creation*: Keep checked.
3. Load Settings o *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.

Load Testing

tejasvivinothkumar2024it / TO DO LIST / Test Plans / Runs / Test run 10 - edit tasks (Ma...

Q Search

⋮ ⏏ ⌚ ⚙️ TV

Completed Test run 10 - edit tasks (Manual)

Run summary Attachments 0 Analytics

100.00% pass rate

✓ 1 Passed

Manual test

Test plan 35 - TO DO LIST

Owned by Tejasvi Vinothkumar

Time completed / duration

10-05-2026 09:30

33s

Pipeline run tested

None

Release & release stage

None

Comment

Edit

Filter by Test case ID or Title

Outcome Configuration Priority Owner

Test results

Bug Link Group by: None

Outcome	Test case	Run by	Priority	Configuration	Owner	Test suite	Iteration	Area
☐ ☒ Passed	45 edit task	Tejasvi Vinot...	2	Windows 10	Tejasvi Vinot...	44 edit tasks	... LIST\sprint 1	TO DO LIST

tejasvivinothkumar2024it / TO DO LIST / Test Plans / Runs / Test run 5 - create task (Ma...

Q Search

⋮ ⏏ ⌚ ⚙️ TV

Completed Test run 5 - create task (Manual)

Run summary Attachments 0 Analytics

100.00% pass rate

✓ 1 Passed

Manual test

Test plan 35 - TO DO LIST

Owned by Tejasvi Vinothkumar

Time completed / duration

10-05-2026 09:20

11s

Pipeline run tested

None

Release & release stage

None

Comment

Edit

Filter by Test case ID or Title

Outcome Configuration Priority Owner

Test results

Bug Link Group by: None

Outcome	Test case	Run by	Priority	Configuration	Owner	Test suite	Iteration	Area
☐ ☒ Passed	38 create task	Tejasvi Vinot...	2	Windows 10	Tejasvi Vinot...	37 create task	... LIST\sprint 1	TO DO LIST

TO DO LIST APPLICATION

Enter Task Title

Enter Description

dd-mm-yyyy 📅

High ▼

Add Task

maths

probability and integral

Due Date: 2026-05-12

Priority: High

Status: Pending

Edit
Complete
Delete

science assignment

physics and chem

Due Date: 2026-05-14

Priority: Medium

Status: Pending

Edit
Complete
Delete

Complete DBMS Assignment

Finish ER diagram and submit report

Due Date: 2026-05-11

Priority: High

Status: Pending

Edit
Complete
Delete

probability and integral

Due Date: 2026-05-12

Priority: High

Status: Pending

Edit
Complete
Delete

physics and chem

Due Date: 2026-05-14

Priority: Medium

Status: Pending

Edit
Complete
Delete

Assignment

Finish ER diagram and submit report

Due Date: 2026-05-11

Priority: High

Status: Pending

Edit
Complete
Delete

Organize Notes

Arrange software engineering notes

Due Date: 2026-05-15

Priority: Low

Status: Pending

Edit
Complete
Delete

water

drink 2L water

Due Date: 2026-05-10

Priority: Medium

Status: Pending

Overdue Task

Edit
Complete
Delete

CAT

study ome

Due Date: 2026-05-11

Priority: Medium

Status: Completed

Edit
Complete
Delete

Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - o In Azure DevOps, create a new project.

- o Create a pipeline and select GitHub as the source.
 - o Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
- o In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - o Add the following basic pipeline configuration for Python and Flask:

yml Code trigger:

- main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

Step 1: Checkout the code from GitHub

- checkout: self

Step 2: Set up Python

environment - task:

UsePythonVersion@0 inputs:

versionSpec: '3.x' # Use the latest Python 3.x version

displayName: "Set up Python"

Step 3: Install dependencies from the correct path

- script: | python -m pip install --upgrade pip pip install -r project/requirements.txt #

Adjusted path to requirements.txt displayName: "Install dependencies"

Step 4: Run a simple Python script to check the environment

- script: | python -c "print('🎵 Hello from Music Playlist Batch Creator!')"

displayName: "Run a Python script"

3. Pipeline Tasks Include:

- o Setting up the Python environment using the UsePythonVersion task.
- o Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- o Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- o Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- o Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

The screenshot displays the Azure DevOps Pipelines interface for a build named "#20260510.1 • Set up CI with Azure Pipelines" under the "TO DO LIST" pipeline. The build status is "Succeeded" (indicated by a green checkmark). The interface includes a search bar at the top right and a sidebar on the left with navigation icons. A notification states: "This run is being retained as one of 3 recent runs by main (Branch)." with a "View retention leases" link. The "Summary" tab is active, showing details for an "Individual CI" build by user "Tejasvi Vinothkumar". The repository is "TO DO LIST" at commit "9151eb51" on the "main" branch. The build started "Today at 14:18" and elapsed "1m 21s". It shows "0 work items" and "0 artifacts". Links for "View 5 changes" and "Parameters" are present. A "Get started" link is also visible under "Tests and coverage". A "Warnings" section at the bottom indicates a deprecation: "Task 'Node.js tool installer' version 0 (NodeTool@0) is deprecated." with a link to "Initialize job".

Result:

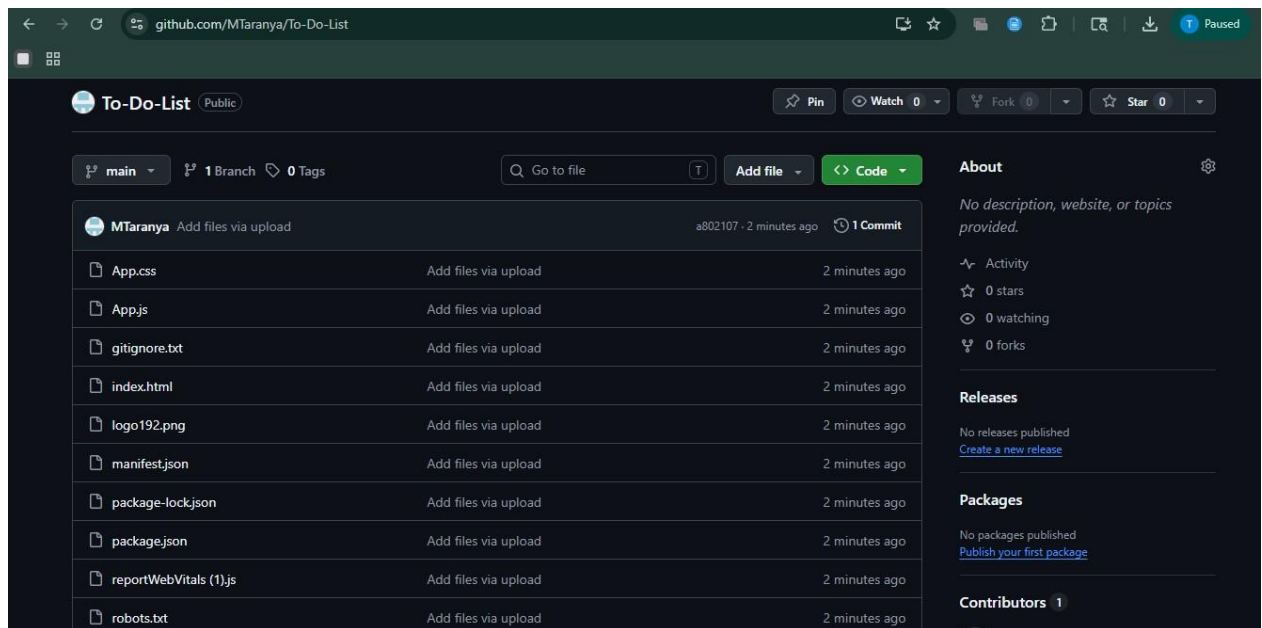
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the To-Do List Application.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.